

SDU 								
Expert in Teams (EXT25)								
Project Name :								
Date :	25/05/2026							
Project Requirements								
REQUIREMENT LIST					REQUIREMENTS ANALYSIS			
ID	REQUIREMENT DESCRIPTION	CATEGORY	PRIORITY	ACCEPTANCE CRITERIA	COMPLEXITY	IN SCOPE?	TEST OR VERIFICATION	REQUIREMENT DELIVERY
<i>Give each requirement a unique ID.</i>	<i>The feature, function, condition or capability that is required from the project.</i>	<i>The category or grouping of the requirement. Example: functional, technical, operational, KPI (result metric), transitional.</i>	<i>The priority of the requirement. Example: P1, P2, P3 or MoSCoW (Must have, should have, could have, won't have).</i>	<i>The criteria that the requirement must have for it to be accepted by the Company & SDU Team</i>	<i>An indication of how complex it will be to deliver the requirement. Example: High, Medium, Low or Very Simple, Simple, hard, not possible, not possible now.</i>	<i>Whether the requirement is going to be in the scope of the project. This will be agreed by the Company & SDU Team.</i>	<i>How the requirement will be checked and tested. For example, it can be based on a performance metric</i>	<i>The project phase or date when the requirement will be delivered (Can be connected to the Gantt chart)</i>
FR-001	Capture a product label image using the device camera.	Functional	Must	User can open capture view and take a photo; image is stored in app storage and linked to a new attempt; user is taken to extraction/processing.	Medium	Yes	Manual acceptance test on iOS and Android using real labels; verify image file exists and attempt is created.	Sprint 2
FR-002	Import a product label image from gallery/files.	Functional	Must	User can select an existing image; image is copied into app storage and linked to a new attempt; extraction/processing starts.	Low	Yes	Manual test on iOS/Android/Web using sample images; verify attempt creation.	Sprint 2
FR-003	Support multi-image capture/import for a single item (split labels).	Functional	Should	User can attach 2+ images to one attempt; extraction uses all images; confirmation screen shows a single merged result with traceability to the source images.	Medium	Yes	Manual test with two images that contain complementary fields; verify merged output contains fields from both.	Sprint 3
FR-004	Persist captured/imported images locally and associate them with attempts.	Functional	Must	Each attempt retains its image(s) across app restarts until deleted by retention rules or user action; storage path is app-private.	Medium	Yes	Restart app and verify attempt still shows the image thumbnail and can re-open images.	Sprint 2
FR-005	Preprocess images before extraction (orientation, resize, basic contrast improvements).	Technical	Should	Preprocessed image(s) are used for extraction; preprocessing never removes readable content (no cropping that loses text); preprocessing can be disabled for troubleshooting.	Medium	Yes	Visual inspection on a small test set; A/B compare extraction accuracy with/without preprocessing on glare/angle samples.	Sprint 2; improved in Sprint 6
FR-006	Decode barcode(s) present on label images.	Functional	Must	For images containing a supported barcode, the app returns the decoded value and symbology; if no barcode is present, extraction continues without failure.	Medium	Yes	Test with sample EAN/UPC/Code128 labels; verify decoded value matches ground truth.	Sprint 2
FR-007	Include decoded barcode values as supplemental data and map them to relevant fields (e.g., EAN/UPC) when applicable.	Functional	Must	Decoded barcode values are stored in the attempt record; if the target schema includes 'ean_or_upc', it is populated from a matching barcode when available; raw barcode list remains available.	Medium	Yes	Integration test on a dataset with known EAN/UPC; verify both mapped field and raw list are present.	Sprint 2

	FR-008	Extract structured data from label images using a schema-driven vision-language (VL) provider (VL-first).	Functional	Must	Given 1+ label images, the provider returns a JSON payload conforming to the internal extraction schema; extraction completes without requiring a separate OCR-only stage.	High	Yes	Provider integration tests using fixed images; validate output JSON against schema.	Sprint 2
	FR-009	Prevent unsupported field inference by limiting what the model is asked to fill and requiring “unknown” for missing evidence.	Technical	Must	Prompts/request schema exclude manual-only fields (e.g., condition); provider instructions require leaving fields blank/null when not explicitly present; confirmations show blanks rather than hallucinated values on labels missing the field.	Medium	Yes	Regression test set where certain fields are absent; verify fields remain empty/null pre-review.	Sprint 3
	FR-010	Validate provider output against a versioned JSON schema and apply normalization rules.	Technical	Must	Output includes ‘schema_version’; runtime validation rejects non-JSON or schema-invalid output; normalization rules are applied (e.g., trim, collapse whitespace; case-insensitive comparisons for evaluation).	Medium	Yes	Unit tests for schema validation and normalization; negative tests for invalid JSON.	Sprint 3
	FR-011	Capture and surface confidence/uncertainty when available.	Functional	Should	If provider supplies per-field confidence, it is stored and shown; if not available, fields are still validated and visually flagged when missing/invalid.	Medium	Yes	Manual test with a provider that returns confidence; verify UI highlights low-confidence fields and allows edits.	Sprint 6
	FR-012	Provide a mandatory confirmation and editing screen prior to submission/export.	Functional	Must	User cannot submit/export without visiting the confirmation screen; invalid/missing required fields are clearly indicated; edited values are used as the final payload.	Medium	Yes	Manual acceptance test: attempt submission is blocked until confirmation; edited value persists in payload.	Sprint 3
	FR-013	Provide dropdown/autocomplete for previously entered values (e.g., Manufacturer) on the confirmation screen.	Functional	Should	Confirmation inputs can suggest previous values from attempt history; selecting a suggestion fills the field; user can still type a new value.	Low	Yes	Manual test: enter manufacturer on attempt A; on attempt B, manufacturer suggestion appears and can be selected.	Sprint 3
	FR-014	Support manual entry of additional registration fields (e.g., item category and condition) and corrections of extracted fields.	Functional	Must	User can enter/override values for all required registration fields; system records whether a field was extracted vs manually set; manual-only fields are never auto-filled by the model.	Medium	Yes	Manual test: set condition and category; verify stored as ‘source=manual’ (or equivalent) and included in submission/export.	Sprint 3
	FR-015	Maintain a mapping layer from internal extraction schema to partner-company registration fields and formats.	Technical	Must	Internal schema remains stable while mapping rules produce an ingestion payload using partner field names/formats; mapping rules are versioned and testable.	High	Yes	Unit tests for mapping rules; acceptance test by posting mapped payload to mock endpoint or validating export file format.	Sprint 1, refined through Sprint 5
	FR-016	Provide configurable provider settings (provider type, endpoint/base URL, model ID).	Functional	Must	User can configure and persist provider settings; extraction uses the configured provider on next run; provider ID/model ID recorded in metadata.	Medium	Yes	Manual test: change provider settings and confirm subsequent extraction uses new provider/model (via metadata).	Sprint 4
	FR-017	Support Bring Your Own Key (BYOK) for remote providers.	Functional	Must	User can enter provider API key/token in Settings; key is required before remote extraction can run; key can be removed/rotated by the user.	Medium	Yes	Manual test: run extraction without key (blocked); add key (succeeds); remove key (blocked).	Sprint 4
	FR-018	Store secrets securely using OS secure storage where supported.	Security	Must	On iOS/Android, secrets are stored via secure storage (Keychain/Keystore); secrets are never written to logs or exported; web/desktop mode shows a warning if secure storage is not available and does not persist secrets by default.	Medium	Yes	Security review + manual inspection (no secrets in local files/logs); platform test on iOS/Android with secure storage APIs.	Sprint 4
	FR-019	Implement provider fallback routing when the primary extraction provider fails.	Functional	Should	If primary provider times out/returns invalid output, the app retries with a configured fallback provider; user is informed which provider succeeded/failed.	High	Yes	Integration test with a forced failure in primary provider; verify fallback succeeds and metadata records provider used.	Sprint 4
	FR-020	Allow users to retry extraction and/or retake images for an attempt.	Functional	Should	User can retry extraction without re-capturing; user can replace an image and re-run extraction; retry count is recorded for the attempt.	Low	Yes	Manual test: retry increments counter; retake image changes image hash and affects extraction.	Sprint 3
	FR-021	Provide configurable ingestion settings (endpoint URL and authentication mode).	Functional	Must	Settings support configuring a POST endpoint URL; supports no-auth and bearer-token auth at minimum; settings persist across restarts.	Medium	Yes	Manual test: configure endpoint and send; verify correct URL and headers.	Sprint 5

	FR-022	Submit confirmed payload via HTTP POST ingestion.	Functional	Must	App sends 'POST' with 'application/json'; any 2xx response marks attempt as 'sent'; non-2xx response records error and leaves attempt queued/unsent.	Medium	Yes	Integration test against mock endpoint returning 2xx and 4xx/5xx; verify status transitions.	Sprint 5
	FR-023	Provide an offline submission queue that stores unsent payloads for later retry.	Functional	Must	If offline or ingestion fails, payload is queued; user can trigger resend (manual or automatic) when back online; queue supports multiple pending items.	High	Yes	Manual test by disabling network; submit; reenable; resend and verify successful POST.	Sprint 5
	FR-024	Ensure the offline queue persists across app restarts.	Functional	Must	Queued payloads remain queued after force close and reopen; queued items are not duplicated; successful sends are removed from the queue.	High	Yes	Reliability test: queue items, force close app 10+ times, verify queue integrity and eventual sending.	Sprint 5
	FR-025	Maintain and display a history of the last 20 attempts with status and details.	Functional	Must	History lists attempts with timestamp and status (e.g., processing, needs review, queued, sent, failed); user can open an attempt to view images, extracted JSON, mapped payload, and errors.	Medium	Yes	Manual test: create >20 attempts and confirm only the latest 20 remain; verify detail view is accessible.	Sprint 3
	FR-026	Enforce retention and deletion rules for attempts.	Operational	Must	The system retains exactly the last 20 attempts and deletes older ones (including related images/payloads); user can manually delete an attempt and optionally clear all local data.	Medium	Yes	Create 21 attempts and verify attempt #1 is deleted; confirm storage usage drops accordingly.	Sprint 3
	FR-027	Support export of confirmed data as JSON and CSV as an integration fallback.	Functional	Must	User can export a confirmed attempt (or batch) to JSON and CSV; exported files match the defined field list and mapping rules; export includes schema version and metadata where applicable.	Medium	Yes	Compare exported files to expected contract format; import test using partner-provided sample importer (if available).	Sprint 5
	FR-028	Provide a local mock ingestion server that simulates the production POST endpoint.	Transitional	Must	Mock server accepts ingestion payload and returns configurable responses (2xx/4xx/5xx); used for development and acceptance testing when partner endpoint is unavailable.	Medium	Yes	Run mock server locally and execute end-to-end submission from the app; verify stored request payloads.	Sprint 5
	FR-029	Record traceability metadata per attempt (timestamp, provider/model IDs, image hash).	Technical	Should	Each attempt stores capture time and a stable identifier (e.g., SHA-256 hash) for each image; provider/model identifiers are stored for audit and evaluation.	Low	Yes	Unit test for hash function; verify metadata is present in stored attempt data.	Sprint 3
	FR-030	Define the minimum target field set to extract and register.	Functional	Must	The internal schema and UI support at minimum: manufacturer, model/type or product number, serial number (when present), country of origin (when present), SKU, product text/description, and EAN/UPC (when present); additional fields (item category, condition) are supported for manual entry.	Medium	Yes	Field checklist review with partner + manual tests on representative labels; verify schema/UI contain these fields.	Sprint 1
	NFR-001	Run on iOS and Android, and provide a usable desktop experience via web (responsive layout).	Technical	Must	Core workflow (capture/import, confirmation/edit, submission/export, history, settings) functions on iOS and Android; web build works in a modern desktop browser for non-camera flows (at minimum import).	High	Yes	Cross-platform test matrix; verify web build import/extraction/confirmation works.	Sprint 2-7, acceptance in Sprint 7
	NFR-002	Remote extraction performance.	KPI	Must	Over a 50 Mbps connection, the confirmation screen is reached within 6.0 seconds for at least 90% of attempts on the reference device.	Medium	Yes	Performance measurement on a fixed image set; report p50/p90 timings.	Sprint 7
	NFR-003	User interface responsiveness.	Non-functional	Must	UI responds to user input within 100 ms for >99% of interactions on the reference device.	Medium	Yes	Manual profiling and instrumentation during UI tests; document results.	Sprint 7
	NFR-004	Cold start time.	Non-functional	Should	Cold start to an operable capture/import screen in < 3.0 seconds on the reference device.	Medium	Yes	Timed cold-start measurements across 10 runs; report median.	Sprint 7
	NFR-005	Offline queue reliability.	Non-functional	Must	Offline queue survives 100% of forced app restarts during a test sequence with pending items.	Medium	Yes	Reliability test script/manual procedure; record pass/fail.	Sprint 5 build; validated in Sprint 7
	NFR-006	Transport security for remote communications.	Security	Must	All remote calls (providers and ingestion) use HTTPS with TLS 1.2+; app rejects non-HTTPS endpoints by default (or requires explicit override with warning).	Medium	Yes	Config validation tests; manual inspection of network traffic in dev mode.	Sprint 4-5

	NFR-007	Data minimization and storage limits per attempt.	Operational	Should	Stored data per attempt (images + JSON + metadata) is kept <= 10 MB on average; large images are resized/compressed to meet this limit without losing readability.	Medium	Yes	Measure storage usage on a sample of attempts; report average and max.	Sprint 6
	NFR-008	Attempt retention limit.	Operational	Must	The app retains exactly the latest 20 attempts and removes older attempts automatically (including files).	Low	Yes	Create 25 attempts and confirm only 20 remain; confirm oldest entries are deleted.	Sprint 3
	NFR-009	End-to-end workflow reliability under stable network conditions.	Non-functional	Must	With stable connectivity, >=95% of attempts complete capture/import -> extraction -> confirmation -> submission without requiring an app restart; failures provide a recovery path (retry, fallback, or queue).	Medium	Yes	Reliability run on a fixed image set with stable network; record completion rate and failure causes.	Sprint 7
	NFR-010	Business-sensitive data handling.	Security	Must	Label images and extracted data are treated as business-sensitive: stored app-private; transmitted only to configured provider(s) and ingestion endpoint; no analytics/telemetry that includes images or extracted fields; users can clear local data.	Medium	Yes	Code/config review + manual test: verify no unexpected network calls and that “clear data” removes local attempts.	Sprint 4-7
	KPI-001	Pre-review extraction accuracy on a held-out dataset.	KPI	Must	For mandatory fields (manufacturer, model/type, serial number, country of origin), achieve >=85% correct pre-review extraction on an agreed held-out set, after normalization rules; report per-field accuracy.	High	Yes	Evaluation harness run on held-out set; publish results table and methodology.	Sprint 6-7
	KPI-002	Registration time reduction compared to manual entry.	KPI	Must	Demonstrate >=50% reduction in median registration time per item using the tool (including confirmation/editing) vs manual entry on the same label set; report median and IQR.	High	Yes	Time study protocol execution; statistical summary in evaluation package.	Sprint 6-7
	KPI-003	Dataset size and split for evaluation.	KPI	Must	Collect >=50 real label images with diverse manufacturers/layouts and capture conditions; split 60% dev and 40% held-out; freeze held-out set for final testing.	Medium	Yes	Dataset inventory file + audit of counts; store split definition and freeze date.	Sprint 1 plan; collection through Sprint 6
	KPI-004	Ground truth quality for evaluation.	KPI	Should	Ground truth transcriptions are created for all evaluation images; a second team member cross-checks a sample and discrepancies are resolved and documented.	Medium	Yes	Review log + spot-check report; inter-annotator discrepancy notes.	Sprint 6
	DOC-001	Deliver D1: partner-approved field list, internal schema, and ingestion/export contract (incl. mapping rules).	Transitional	Must	Field list and formats are reviewed with partner; schema is versioned; ingestion contract specifies endpoint/auth/payload; mapping rules documented.	Medium	Yes	Document review session with partner; mock endpoint accepts payload matching the contract.	Sprint 1 and Sprint 5 finalization
	DOC-002	Deliver D2: functional prototype implementing the end-to-end workflow.	Transitional	Must	Prototype supports capture/import, extraction, confirmation/edit, submission/queueing, settings, and history on iOS/Android; demo flow reproducible.	High	Yes	End-to-end demo run using representative labels; acceptance checklist signed by team.	Sprint 5
	DOC-003	Deliver D3: evaluation package (dataset description, methodology, and results).	Transitional	Must	Package includes dataset description, ground truth procedure, per-field accuracy, end-to-end success rate, and time-savings results.	Medium	Yes	Re-run evaluation from documented steps and reproduce results.	Sprint 7
	DOC-004	Deliver D4: technical documentation (install, configuration, security model, troubleshooting).	Transitional	Must	A new developer can install and run the prototype; configuration steps for providers/ingestion are documented; security assumptions and limitations are stated.	Low	Yes	Documentation walkthrough by a team member not authoring the doc; fix identified gaps.	Sprint 8
	DOC-005	Deliver D5: final demo and handover package.	Transitional	Must	Includes demo script, known limitations, and a prioritized backlog for future work; final presentation materials prepared.	Low	Yes	Dry-run demo and presentation; verify handover artifacts are complete.	Sprint 8
	D-001	Local vision-language (VL) extraction mode (on-device).	Technical	Could	A supported device can run extraction without network access; output remains schema-valid; user can switch between local and remote extraction.	High	Yes (time permitting)	Benchmark on-device extraction on a small test set; compare accuracy/latency vs remote.	Sprint 6-8
	D-002	Provider benchmarking and comparison tooling.	KPI	Could	Tooling can run the same dataset through multiple providers/models and produce comparable metrics (accuracy, latency, cost notes).	Medium	Yes (time permitting)	Repeatable benchmark run documented; results table produced.	Sprint 6-7

	D-003	Confidence-focused user interface enhancements.	UX	Could	Confirmation UI clearly highlights low-confidence fields (where available) and missing/invalid fields; user can filter/jump to fields needing attention.	Medium	Yes (time permitting)	Usability walkthrough on a test set; measure time-to-correct vs baseline UI.	Sprint 6-7
	D-004	Image/keyword search helper for manufacturer or product info (agentic/assisted lookup).	Functional	Could	User can search by image or keywords to retrieve reference info (e.g., manufacturer name normalization); any retrieved data is explicitly marked as “suggested” and requires user confirmation.	High	Yes (time permitting)	Demo with controlled examples; verify suggested values never auto-submit without confirmation.	Sprint 7-8
	D-005	Support additional local VL models for image-to-JSON extraction.	Technical	Could	App can download/select at least one additional local model and run extraction with the same schema; model selection is exposed in Settings.	High	Yes (time permitting)	Manual test on supported devices; verify schema validation passes across models.	Sprint 7-8
	CON-001	Do not train custom vision models on proprietary company data.	Constraint	Must	No custom training pipeline or trained proprietary model artifacts are produced; solution uses existing local/remote models only.	Low	Yes	Architecture/code review; confirm evaluation uses off-the-shelf models.	Throughout
	CON-002	Do not implement full integration into the partner’s internal UI; deliver ingestion contract + adapter/export instead.	Constraint	Must	No changes are required in the partner UI to test the prototype; integration is demonstrated via configurable HTTP ingestion and/or export files.	Low	Yes	Acceptance test using mock endpoint and export files; partner review of contract.	Sprint 5
	CON-003	Do not infer fields that are not present on the label (e.g., condition) beyond user entry.	Constraint	Must	Extracted outputs do not include invented values for non-label fields; such fields remain empty until user provides them.	Medium	Yes	Regression tests with missing fields; manual spot checks on evaluation set.	Sprint 3
	CON-004	Do not target large-scale production deployment or enterprise device management.	Constraint	Won't	Deliverable is a functional prototype and evaluation; no MDM, SSO, or production rollout plan beyond a handover backlog.	Low	Yes	Review deliverables vs scope boundaries.	Sprint 8

DEFINITIONS

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